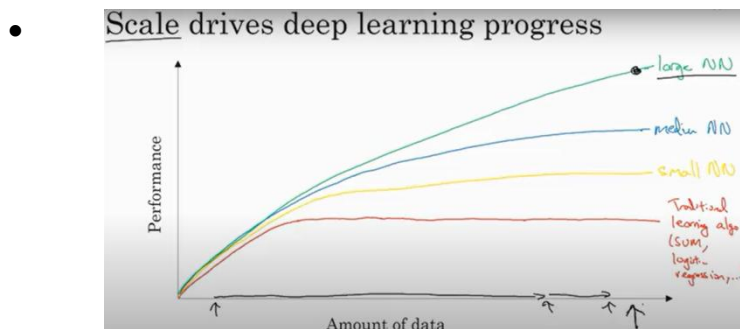


Course 1: Neural Network & Deep Learning

- What is a Neural Network?

It basically acts like a function taking in an input, x and generating an output, y based on x . The input may include a single parameter or a combination of many parameters ($x_1, x_2, x_3, \dots$). It consists of neurons which take decisions and lead to an output.



Larger NNs perform better at larger amount of data while smaller NNs tend to saturate after some point.

- Logistic Regression - $\hat{y} = \sigma(w^T x + b)$, where $\sigma(z) = \frac{1}{1+e^{-z}}$

Loss Function $\mathcal{L}(\hat{y}, y) = - (y \log \hat{y} + (1-y) \log(1-\hat{y})) -$

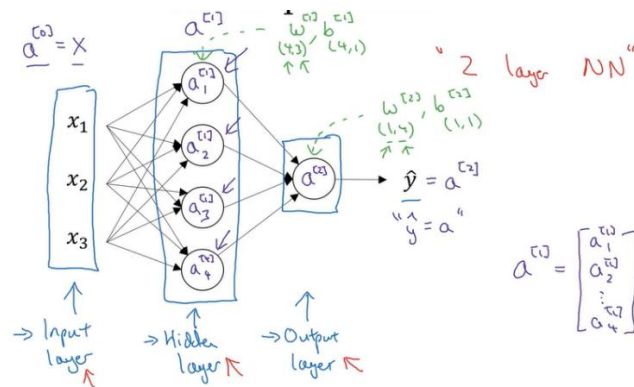
Cost function is the average of loss function summed over all the data-points.

- Gradient Descent:

$$w := w - \alpha \frac{\partial J(w, b)}{\partial w}$$
$$b := b - \alpha \frac{\partial J(w, b)}{\partial b}$$

- **Vectorisation** – It can be used in place of for loops to make the code a lot faster and simpler.

- **Broadcasting** in Python can change the dimension of matrix implicitly to suit the matrix calculations.
- Use strictly row/column vectors instead of rank 1 arrays to resolve vector-size inconsistency.
- NN Representation

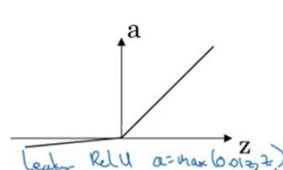
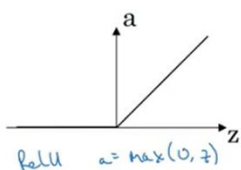
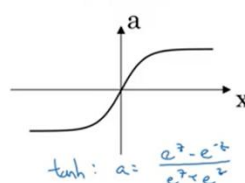
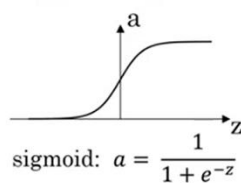


'a' is the Activation or the value of a particular node which is to be passed to the next layer.

- Vectorizing across multiple examples:

```
for i = 1 to m:
     $z^{[1]}(i) = W^{[1]}x^{(i)} + b^{[1]}$ 
     $a^{[1]}(i) = \sigma(z^{[1]}(i))$ 
     $z^{[2]}(i) = W^{[2]}a^{[1]}(i) + b^{[2]}$ 
     $a^{[2]}(i) = \sigma(z^{[2]}(i))$ 
```

- Activation Functions:



- Important Formula:

Forward propagation:

$$z^{[1]} = w^{[1]}x + b^{[1]}$$

$$A^{[1]} = g^{[1]}(z^{[1]}) \leftarrow$$

$$z^{[2]} = w^{[2]}A^{[1]} + b^{[2]}$$

$$A^{[2]} = g^{[2]}(z^{[2]}) = \sigma(z^{[2]})$$

Back propagation:

$$dz^{[2]} = a^{[2]} - y$$

$$dW^{[2]} = dz^{[2]}a^{[1]T}$$

$$db^{[2]} = dz^{[2]}$$

$$dz^{[1]} = W^{[2]T} dz^{[2]} * g^{[1]'}(z^{[1]}) \quad dz^{[1]} = W^{[2]T} dz^{[2]} * g^{[1]'}(z^{[1]})$$

$$dW^{[1]} = dz^{[1]}x^T$$

$$db^{[1]} = dz^{[1]}$$

$$dZ^{[2]} = A^{[2]} - Y$$

$$dW^{[2]} = \frac{1}{m} dZ^{[2]} A^{[1]T}$$

$$db^{[2]} = \frac{1}{m} \text{np.sum}(dZ^{[2]}, \text{axis} = 1, \text{keepdims} = \text{True})$$

$$dZ^{[1]} = W^{[2]T} dZ^{[2]} * g^{[1]'}(Z^{[1]})$$

- Dimensions for a Deep NN:

$$W^{[2]}: (n^{[2]}, n^{[1]})$$

$$b^{[2]}: (n^{[2]}, 1)$$

- Parameters include weights and biases.

Hyper-parameters include learning rate(α), iterations, number of hidden layers, etc.